Reward Recovery in Sequential Decision Models

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1 Setup and Notation

Finite state space $\mathcal{S}$, finite action space $\mathcal{A}$, discount factor $\beta \in (0,1)$, transition kernel $P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, unknown reward $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$. Additive T1EV shocks yield the soft Bellman recursion:

$$Q^*(s,a) = r(s,a) + \beta \sum_{s'} P(s' \mid s,a)\, V^*(s'), \tag{1}$$

$$V^*(s) = \log \sum_{a'} \exp Q^*(s,a'), \tag{2}$$

$$\pi^*(a \mid s) = \exp\big(Q^*(s,a) - V^*(s)\big), \tag{3}$$

$$A^*(s,a) = Q^*(s,a) - V^*(s) = \log \pi^*(a \mid s). \tag{4}$$

The advantage A^* is identified from CCPs. The reward r is not: any $\tilde{r}(s,a) = r(s,a) + \lambda(s)$ generates identical CCPs (Magnac and Thesmar, 2002). Following Kalouptsidi et al. (2021), call the set of all such $\tilde{r}$ the identified set.

2 Four Approaches to Reward Recovery

2.1 Reduced-Form Q

Estimate $Q_\omega(s,a)$ by logit MLE:

$$\hat{\omega} = \arg\max_{\omega} \sum_{i,t} \log \frac{\exp Q_\omega(s_{it}, a_{it})}{\sum_{a'} \exp Q_\omega(s_{it}, a')}. \tag{5}$$

At the optimum, $Q_{\hat{\omega}}(s,a) = Q^*(s,a) + c(s)$ for action-independent c. The backed-out "reward" is $A^*(s,a)$, not $r(s,a)$.

2.2 AIRL (Fu et al., 2018)

The discriminator decomposes as

$$f_\theta(s,a,s') = g_\theta(s,a) + \beta\, h_\phi(s') - h_\phi(s), \tag{6}$$

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where g_θ approximates the reward and $\beta h(s') - h(s)$ is potential-based shaping (Ng et al., 1999). At the GAN optimum, $f^* = A^*$ (Fu et al., 2018, Appendix A.4), so

$$g^*(s,a) = r(s,a) - \delta(s) + \beta \sum_{s'} P(s' \mid s,a)\, \delta(s'), \tag{7}$$

where $\delta(s) \equiv V^*(s) - h^*(s)$. Unless $\delta \equiv 0$, the recovered g^* is a potential-based perturbation of r.

2.2.1 LSW Anchor Identification (Lee et al., 2026)

Two anchors: (A1) $g(s, \text{exit}) = 0$ for all s; (A2) $h(s_{\text{abs}}) = 0$, where $P(s_{\text{abs}} \mid s, \text{exit}) = 1$.

Lemma 1 (Non-identifiability)**.** *If (g,h) and $(\bar{g},\bar{h})$ produce the same f for all (s,a,s'), then $\bar{h} = h + \Phi$ and $\bar{g}(s,a) = g(s,a) + \Phi(s) - \beta\Phi(s')$ for some $\Phi : \mathcal{S} \to \mathbb{R}$.*

Proof. Rearrange $g + \beta h(s') - h(s) = \bar{g} + \beta\bar{h}(s') - \bar{h}(s)$ and define $\Phi = \bar{h} - h$. □ □

Theorem 2 (Anchor identification)**.** *Under (A1)–(A2), if (g,h) and $(\bar{g},\bar{h})$ realize the same f, then $g = \bar{g}$ and $h = \bar{h}$.*

Proof. By Lemma 1, $\bar{g}(s, \text{exit}) = 0 + \Phi(s) - \beta\Phi(s_{\text{abs}})$. Anchor (A1) forces $\Phi(s) = \beta\Phi(s_{\text{abs}})$ for all s. Anchor (A2) forces $\Phi(s_{\text{abs}}) = 0$. Hence $\Phi \equiv 0$. □ □

Theorem 3 (Recovery of true reward)**.** *If the true pair (r^*, V^*) satisfies $r^*(s, \text{exit}) = 0$ and $V^*(s_{\text{abs}}) = 0$, then any AIRL pair (g,h) realizing A^* under the same anchors satisfies $g = r^*$ and $h = V^*$.*

Proof. Define $\Phi = h - V^*$. The exit anchor gives $\Phi(s) = \beta\Phi(s_{\text{abs}})$. The absorbing anchor applied to both h and V^* gives $\Phi(s_{\text{abs}}) = 0$. □ □

The Kang et al. (2025) ERM framework uses the same anchor principle (fixing the reward of a known action) but solves a single non-adversarial optimization.

2.3 IQ-Learn (Garg et al., 2021)

The inverse Bellman operator $r_Q(s,a) \equiv Q(s,a) - \beta \sum_{s'} P(s' \mid s,a)\, V_Q(s')$ expresses reward as a function of Q alone. IQ-Learn estimates Q by

$$\max_Q \; \mathbb{E}_{(s,a)\sim\rho^*}\big[\phi\big(r_Q(s,a)\big)\big] - (1-\beta)\, \mathbb{E}_{s_0\sim\mu_0}\big[V_Q(s_0)\big], \tag{8}$$

where $\phi(x) = x - x^2/2$ for χ^2. At the optimum, r_Q separates immediate reward from continuation values via the Bellman constraint.

2.4 GLADIUS (Kang et al., 2025)

Parameterizes $Q(s,a)$ and $\hat{\mathbb{E}}[V(s') \mid s,a]$ with neural networks. Trains via logit NLL plus Bellman consistency penalty on observed (s,a,s') transitions. Extracts reward via the same inverse Bellman operator as IQ-Learn. The MLP architecture shares parameters across states, providing implicit regularization over tabular Q.

3 Counterfactual Taxonomy

Definition 4 (Type I: state extrapolation)**.** *P and r unchanged; only realized state values shift. Counterfactual CCPs require only evaluating the estimated Q at new inputs.*

Definition 5 (Type II: environment change)**.** *r unchanged but $P \to \tilde{P}$ or the action set changes. Counterfactual Q must satisfy a new Bellman equation $\tilde{Q} = r + \beta \tilde{P} \tilde{V}$. Requires r separated from continuation values.*

Proposition 6 (Type I equivalence)**.** *Under Type I, all four methods yield identical counterfactual CCPs.*

Proof. Each recovers $Q^*(s, a) + c(s)$ for some action-independent c. The softmax $\pi(a \mid \tilde{s}) \propto \exp(Q^*(\tilde{s}, a) + c(\tilde{s}))$ cancels c. □ □

Proposition 7 (Type II divergence)**.** *Under Type II, reduced-form produces invalid predictions. AIRL with anchors, IQ-Learn, and GLADIUS produce valid predictions when r is correctly recovered.*

The reduced-form estimate $Q^* + c(s)$ bundles r with $\beta P V^*$. Under $\tilde{P}$, continuation values change but the reduced-form has no way to disentangle r. AIRL with anchors recovers $g^* = r^*$ exactly (Theorem 3). IQ-Learn and GLADIUS separate r via the inverse Bellman operator. The LSW anchors provide what Kalouptsidi et al. (2021) call the "additional restrictions" needed to identify counterfactuals under transition changes.

4 Application to Lee, Sudhir & Wang (2026)

LSW study sequential content consumption (buy, wait, or exit at each episode) with AIRL under anchors (A1)–(A2) and deterministic transitions. Their counterfactuals classify as:

- Wait-time validation (Section 6.3): Type I. A reduced-form logit passes this test identically.
- WFF removal and segment-specific delays (Sections 7.1–7.2): Type II. These change the action-set structure and the MDP each segment faces.
- Content-based paywall (Section 7.3): Type II. High-content episodes become paid-only, altering transitions episode-by-episode.

5 Simulation Study

Tabular MDP mimicking LSW: 15 episodes × 3 content levels × 4 wait-times = 180 regular states plus one absorbing state. Actions $\in$ {buy, wait, exit}, deterministic transitions, $\beta = 0.95$. True reward: $r^*(s, \text{buy}) = u(c) - 2.0$, $r^*(s, \text{wait}) = u(c) - 0.08w$, $r^*(s, \text{exit}) = 0$, where $u \in \{0.5, 1.5, 3.0\}$. Analyses A–C and E–F are population-level (no estimation noise). Analysis D uses finite-sample estimation.

5.1 A. Reward recovery

5.2 B. Type II error vs. transition shift

Buying advances k episodes instead of 1. Bellman re-solved under each method's reward.

Method	MSE vs. r^*	Corr. with r^*
Oracle / AIRL+anchors / IQ / GLADIUS	0.000	1.000
AIRL-no-anchors	10.005	0.440
Reduced-form	40.020	0.138

Table 1: Population-level reward recovery. Type I CCP errors are below 10^{-6} for all methods.

k	Oracle/AIRL+a/IQ/GLADIUS	AIRL-no-a	Reduced-form
1	0.000	0.000	0.000
2	0.000	0.025	0.082
3	0.000	0.017	0.094
5	0.000	0.015	0.123
7	0.000	0.016	0.129
10	0.000	0.014	0.139

Table 2: Type II CCP error. Structural methods are exact. Reduced-form error grows with shift magnitude.

5.3 C. Shaping magnitude sweep

Shaped rewards $\tilde{r} = r^* + \alpha V^*(s) - \alpha\beta V^*(s')$, $\alpha \in [0, 1]$. Type II CF with $k = 3$.

5.4 D. Finite-sample convergence

Tabular χ^2 IQ-Learn has multiple local optima in finite samples. We compare: (i) reduced-form logit (converges to A^*, wrong for Type II), (ii) anchor-constrained MLE (logit NLL + anchor penalties on implied reward), (iii) GLADIUS (neural Q + EV).

5.5 E. Anchor misspecification

True exit payoff $= \varepsilon$, anchors assume 0.

5.6 F. Content-based paywall

High-content episodes become paid-only (wait blocked). This is a Type II shift.

6 Summary

References

Fu, J., Luo, K., and Levine, S. (2018). "Learning Robust Rewards with Adversarial Inverse Reinforcement Learning." ICLR.

Garg, D., Chakraborty, S., Cundy, C., Song, J., and Ermon, S. (2021). "IQ-Learn: Inverse soft-Q Learning for Imitation." NeurIPS.

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α	Type II CF error	Corr. with r^*
0.00	0.000	1.000
0.10	0.004	0.908
0.30	0.011	0.610
0.50	0.017	0.440
1.00	0.027	0.270

Table 3: CF error is monotone in shaping magnitude. Without anchors, α is indeterminate.

N	Reduced-form	Anchor-MLE	GLADIUS
500	0.264	0.255	0.106
2,000	0.262	0.255	0.123
10,000	0.262	0.257	0.099

Table 4: Type II CF error ($k = 3$, 2 seeds). GLADIUS cuts error by 60% over tabular methods via implicit neural regularization.

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ε	Corr. with r^*	Type II CF error	In-sample π shift
0.0	1.000	0.000	0.000
0.5	0.976	0.002	0.000
1.0	0.909	0.004	0.000
2.0	0.729	0.008	0.000

Table 5: Degradation is smooth, not knife-edge. In-sample CCPs are unaffected.

Method	CCP error
Oracle / AIRL+anchors / IQ-Learn	0.000
AIRL-no-anchors	0.009
Reduced-form	0.185

Table 6: Reduced-form fails at 18.5pp because it cannot separate content utility from continuation value.

	Reduced-form	AIRL (generic)	AIRL (anchors)	IQ-Learn / GLADIUS
Recovers	$A^*(s,a)$	r + shaping	$r^*(s,a)$	$r^*(s,a)$
Type I CF	✓	✓	✓	✓
Type II CF	×	Only if $\delta \equiv 0$	✓	✓